

Subject: 19CSE456

Lab Session: 06

Notes:

1. Please read the assignment notes carefully and comply to the guidelines provided.
2. A report should be prepared for questions available in “Report Section” of this assignment & submitted to TurnItIn. Only one member of the team should upload the report to TurnItIn. The submission title should start with the team name / number.
3. The report should not have any content (text, figures, photos etc.) copied from any source outside your work. Please provide results obtained from your experiment to support your statements.
4. Any content retrieved from any external source (papers, blogs, websites, chat sites etc.) should be duly acknowledged.

References:

- <https://towardsdatascience.com/convolutional-neural-network-feature-map-and-filter-visualization-f75012a5a49c>
- <https://valueml.com/get-the-output-of-each-layer-in-keras/>

Main Section (Mandatory):

A1. Use CNN (standard architectures as well as your custom designed architectures for your project. If your project deals with generation of images, learn to use CNN for generative purposes.

A2. For classification & regression projects, learn to take the feature map from the layer before the output of the CNN. Use the extracted feature vector for classification and regression using standard classifiers & regressors.

- Suggested Classifiers – kNN, Decision Tree, Random Forest, SVM, Naïve-Bayes classifier etc.
- Suggested Regressors -- Linear Regression, LASSO, k-NN, SVR, Decision Tree, Random Forest, ADA Boosted Trees, Gradient Boosted Trees, XG Boosting Trees etc.

A4. Learn & use appropriate metric to report your generative results.

Report Assignment:

1. Update your last week's report with the study of classification using CNN. Fill all relevant sections for a paper. [4]